

Project Details

Group M

Members	Roll Number
Sanket Prakash	230103121
Rahul Yadav	230103123
Vineet Agarwal	230103116

Hydraulic Jack

Project Description:

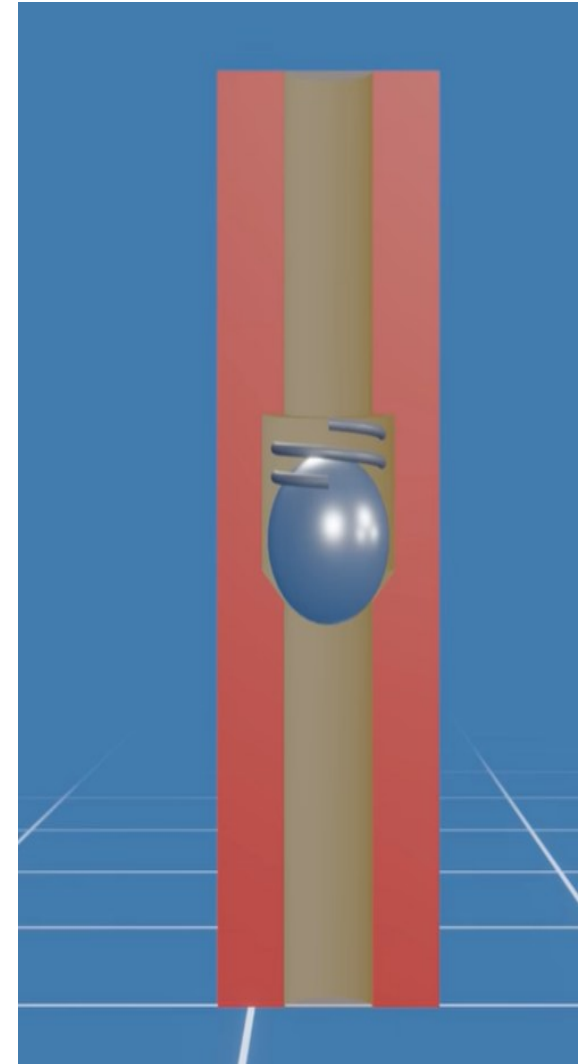
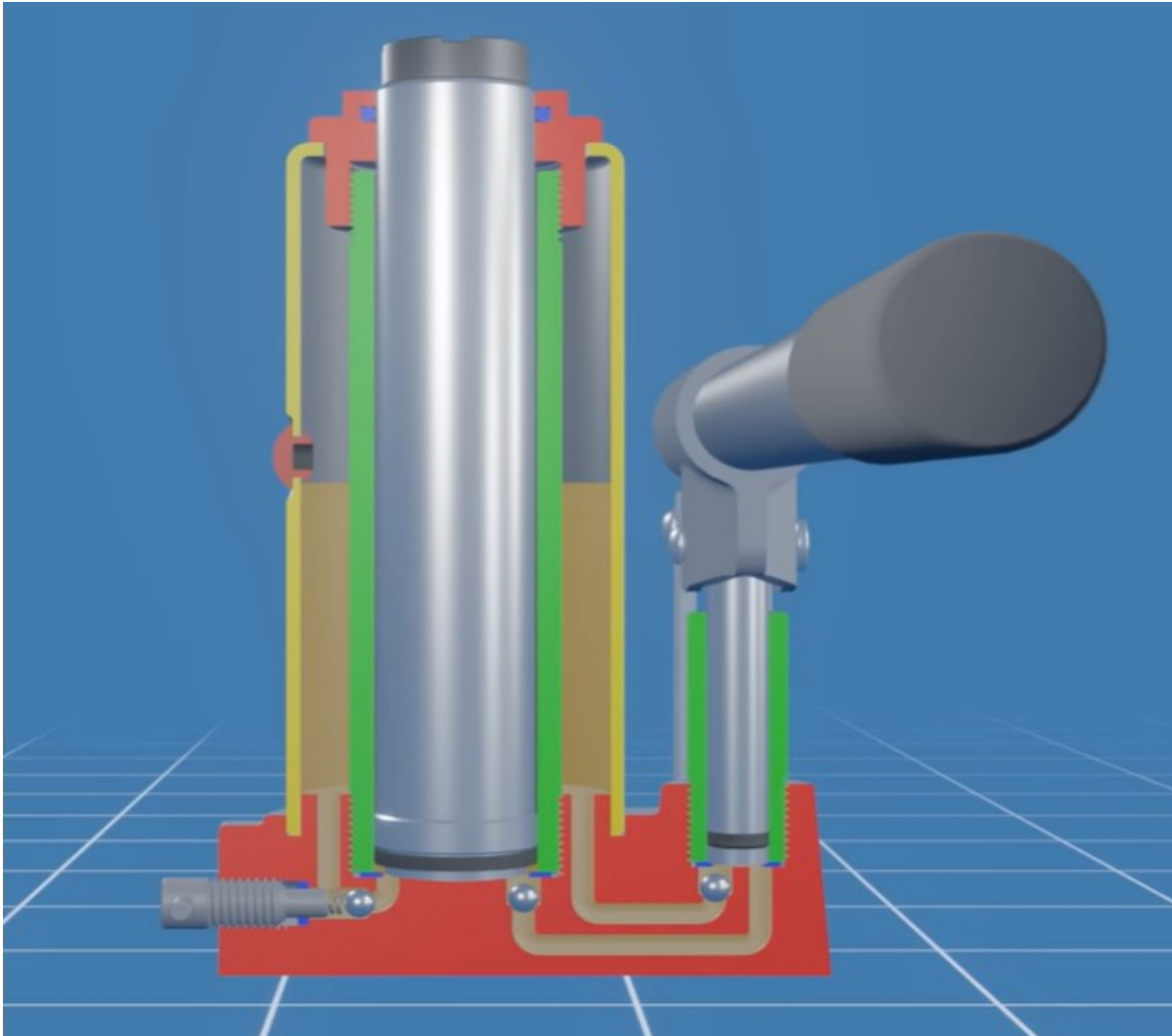
- Basic idea is to analyse a hydraulic bottle jack, including the details of the material.
- Analysis will be based on details of an actual hydraulic bottle jack, namely Taparia HBJ3.
- Firstly, the complete anatomy and morphology of the jack will be studied, and then we will try to collect the details about the basic philosophy of design.
- Ultimately, we will try to find a novel idea for its further development and look into its implementation details.



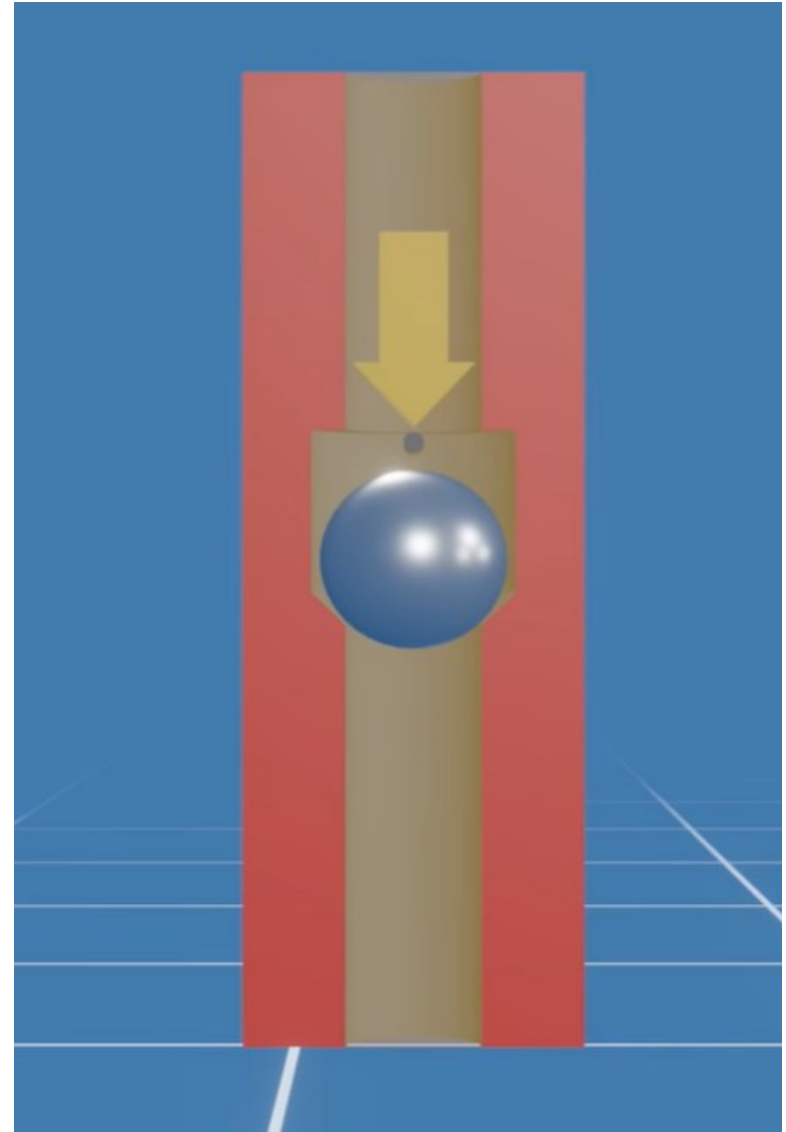
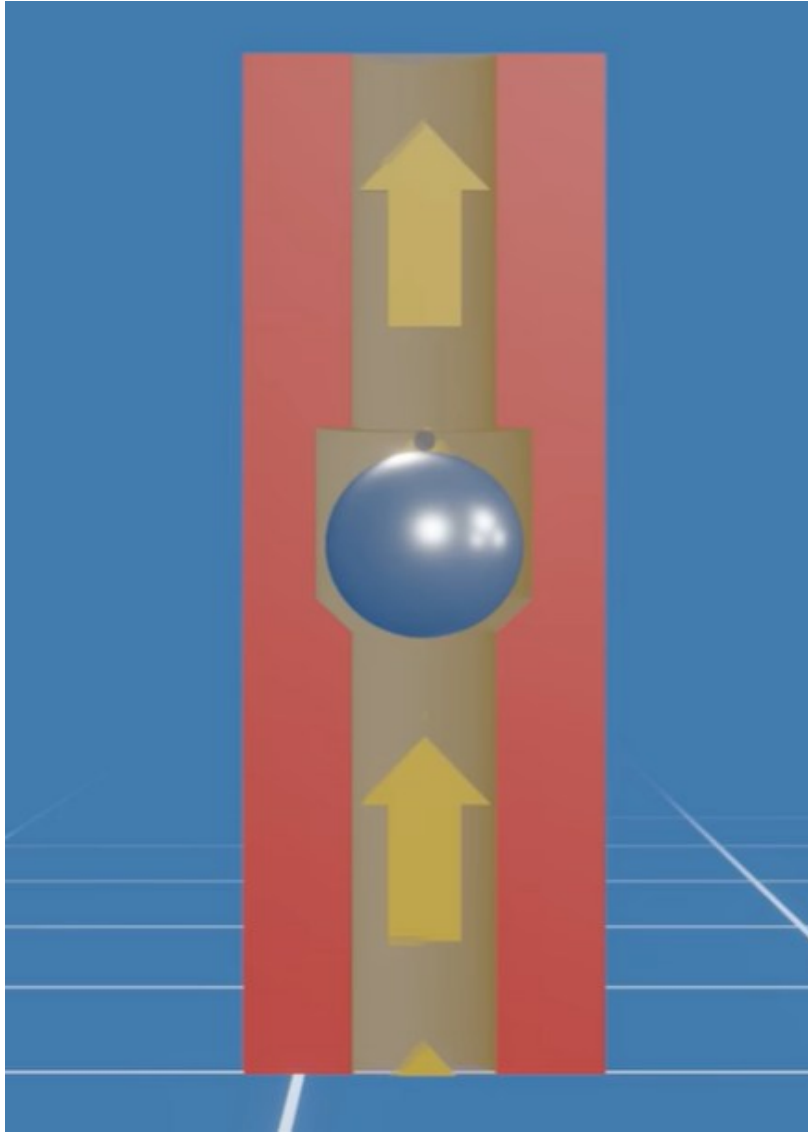
Working of an HBJ

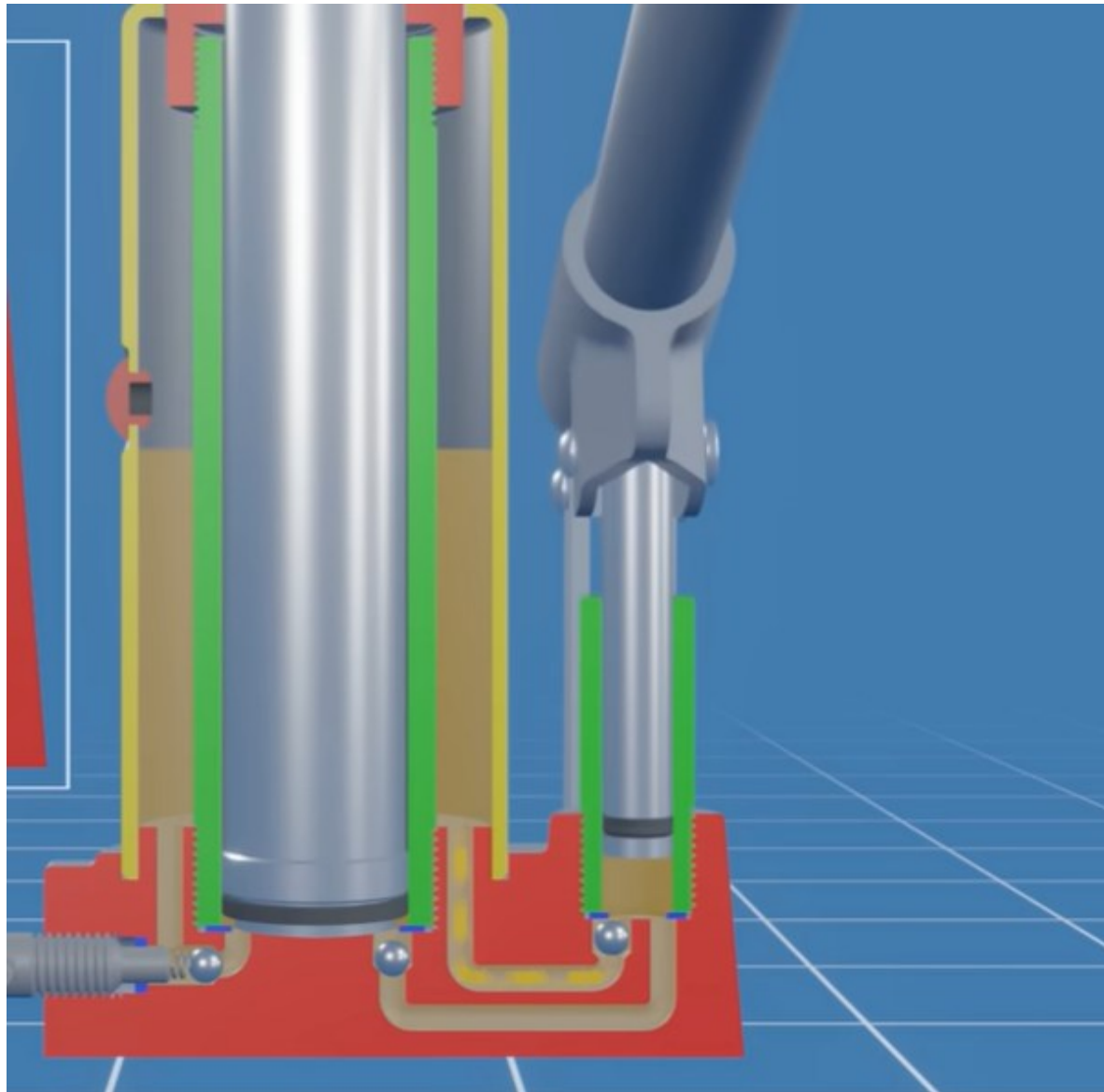
DIAGRAMS INDICATING
FLOW

Working of an HBJ



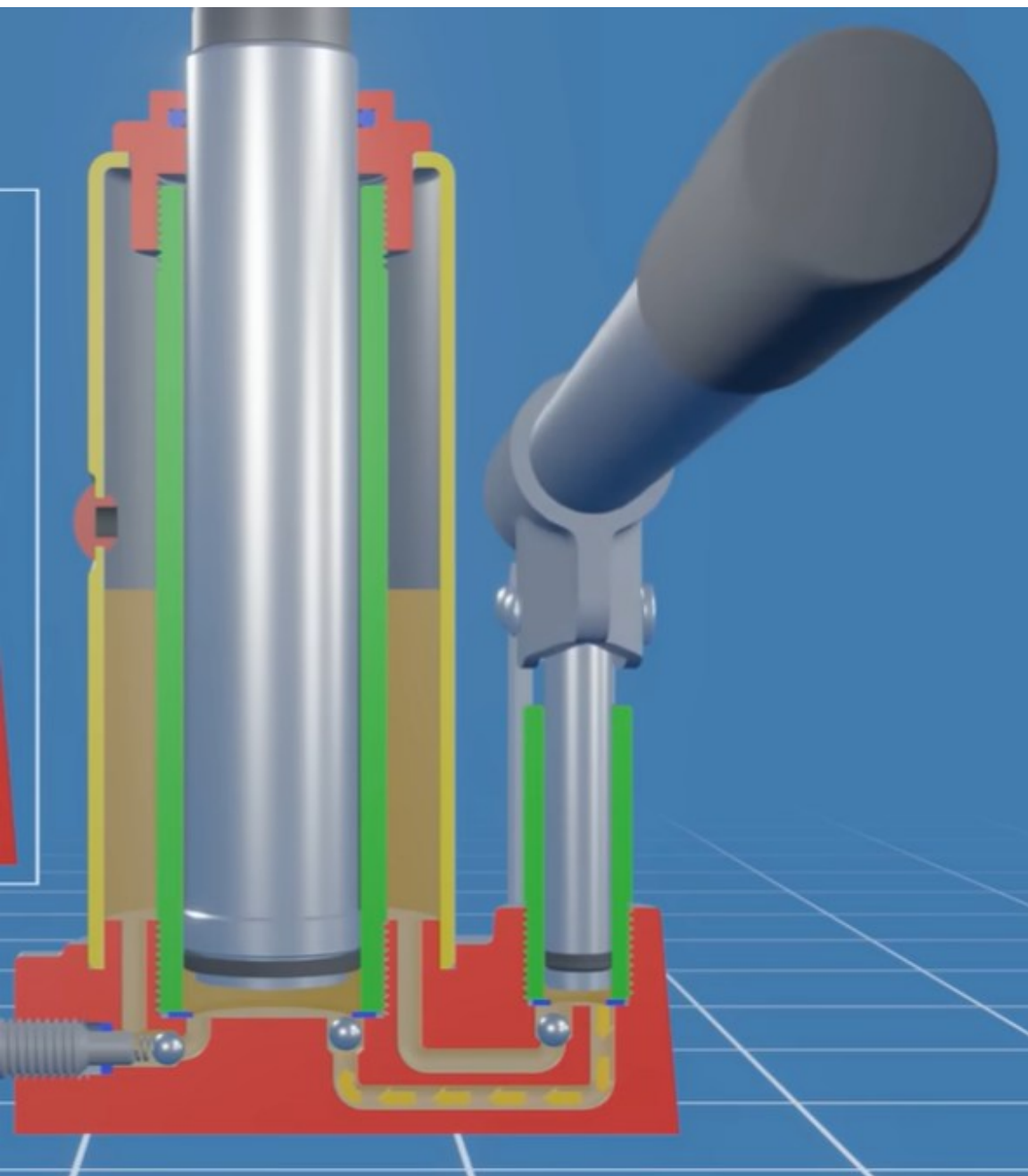
Oil Flow and Ball Movement





When pump piston moves up, the hydraulic oil is sucked from the oil reservoir to the pump cylinder through an inlet passage.

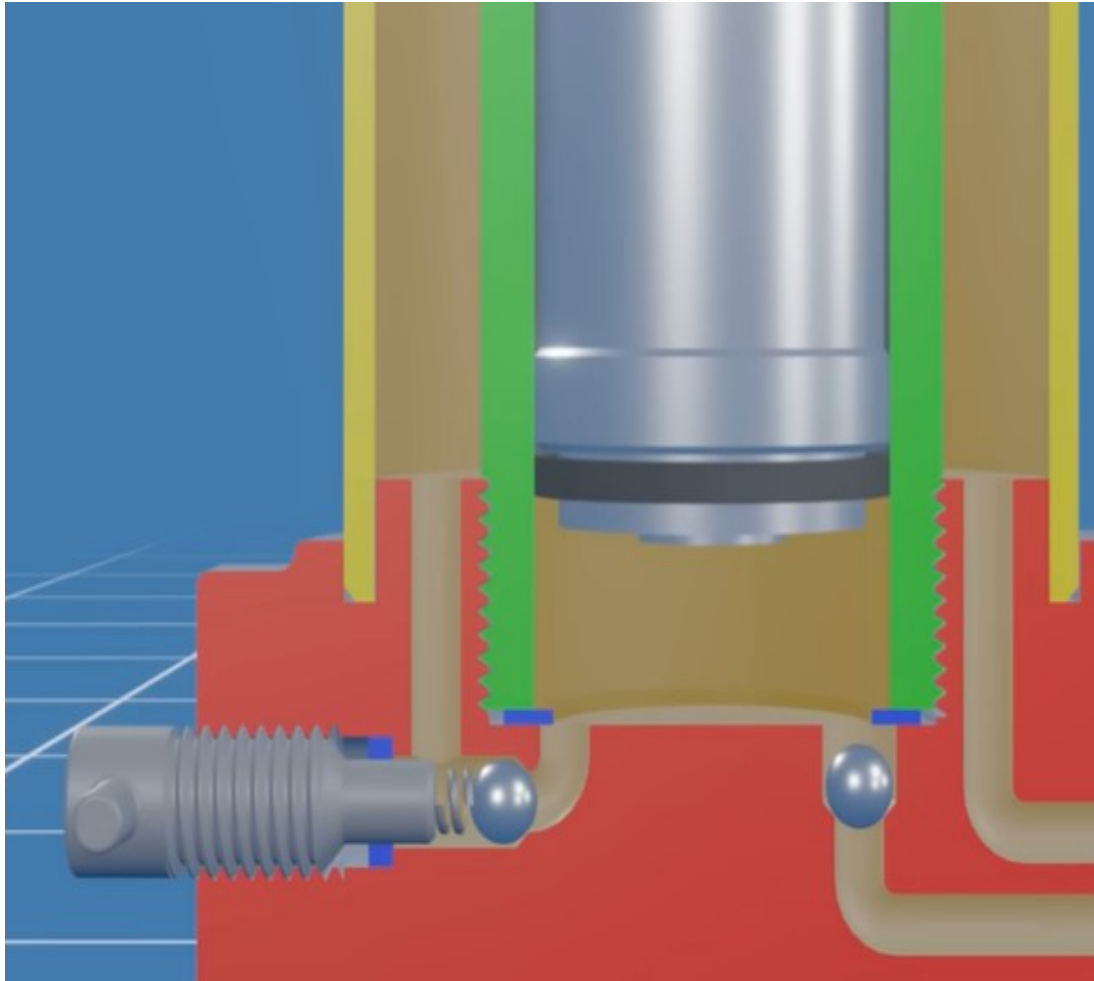
Just before entering the pump cylinder, the fluid lifts the steel ball allowing the fluid to pass through it freely.



As the pump piston subsequently moves down, it pressurises the hydraulic oil, pushing this steel ball down, closing up the pathway back to the reservoir.

And, forcing the hydraulic oil through the second passage of the main cylinder.

Simultaneously, the steel ball in the main cylinder is lifted by the incoming pressurised oil, allowing the fluid to flow in; it pushes against the piston, causing it to rise.



To lower the load =>
Rotate release valve
ACW

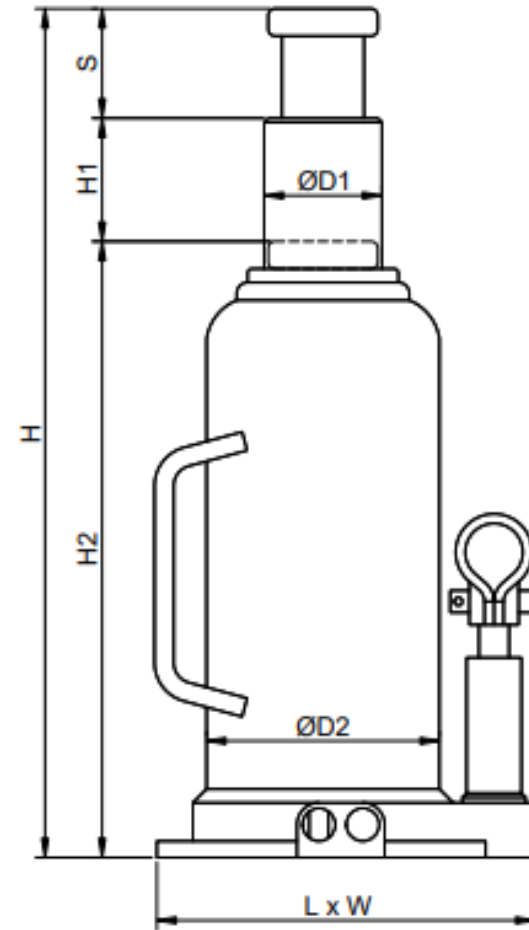
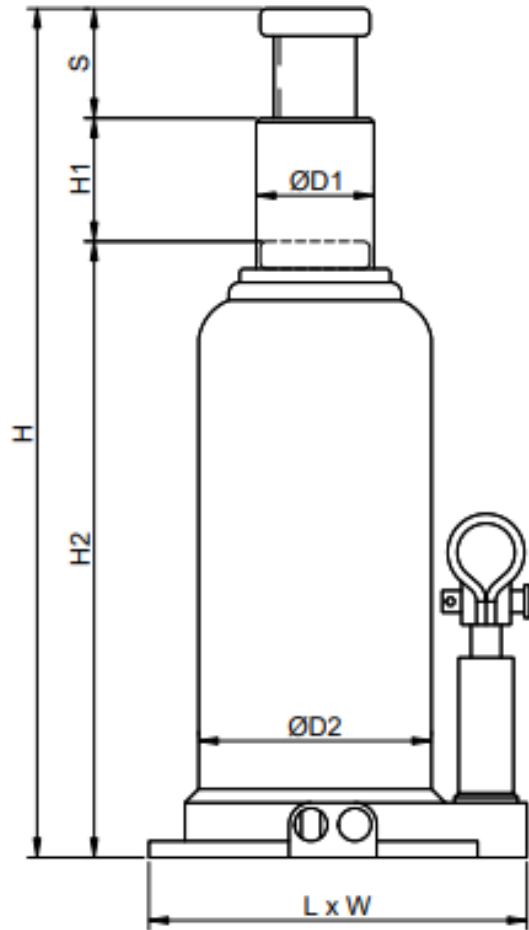
Details of Taparia HBJ3

BASED ON THE
INFORMATION BY
MANUFACTURER



TAPARIA
AN ISO - 9001 COMPANY

Physical Dimensions

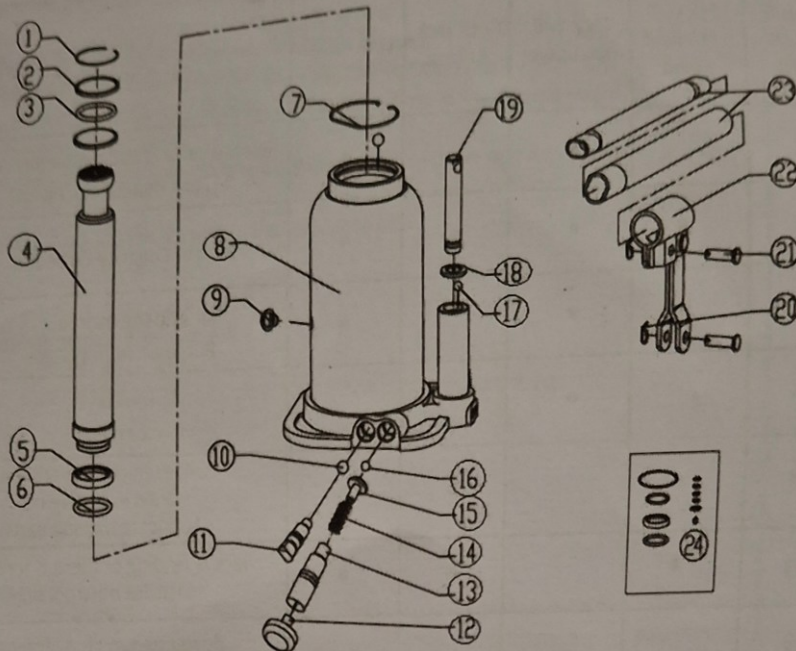


Physical Dimensions

PRODUCT NO.	FIGURE NO.	CAPACITY IN TONNES	L X W BASE SIZE	H TOTAL HEIGHT	H1 HYDRAULIC LIFT	H2 CLOSED HEIGHT	S SCREW EXTENSION	ØD1 RAM DIA	ØD2 RAM DIA	WEIGHT (KG)
HBJ 2	1	2	82X5.5	338	112	178	48	22	48	2.08
HBJ 3		3	94X5	363	115	188	60	28	60	2.82
HBJ 5		5	100X6	402	125	207	70	32	70	3.72
HBJ 8		8	110X6	447	145	222	80	35	75	4.70
HBJ 10		10	110X6	447	145	222	80	35	75	4.70
HBJ 12		12	120X6	457	150	227	80	43	85	6.00
HBJ 16		16	125X7	457	150	227	80	48	96	6.97
HBJ 20		20	135X7.5	445	150	235	60	53	110	8.78

Assembly Diagram

Assembly Diagram:



Part No	Description	Qty
1	Round wire snap ring	1
2	Ring (Rectangular Section)	2
3	O-Ring	1
4	Ram assembly	1
5	Bowl washer	1
6	O-Ring	1
7	Round wire snap ring	1
8	Base welding parts	1
9	Oil plug	1
10	Steel ball 6 mm	1
11	Release valve assembly	1
12	Plastic plug	1
13	Safety valve	1
14	Spring	1
15	Spring holder	1
16	Steel ball 3 mm	1
17	Steel ball 5 mm	2
18	Bowl washer	1
19	Pump plunger assembly	1
20	Reusable cotter pin 01.2 x 18 mm	2
21	Pin	2
22	Handle socket assembly	1
23	handle assembly	1
24	Seal kit	1

Salient Features

SALIENT FEATURES

Bottle jack is made from high grade steel.

Bottle jack is a vertical hydraulic jack.

Compact and rigid design which occupies less space and easy to handle.

Superior hydraulic fluid is used.

Sturdy steel base structure to avoid fluid leakages

it works in all weather conditions.

Bottle jacks are powder coated to prevent corrosion

Vide application in automobile garages / work stations, cars, trucks and industries etc.



Designing an HBJ

BASED ON [TBEBU, PETROS. "DESIGN
OF MECHANICAL HYDRAULIC JACK,"
N.D. DOI:10.9790/3021-04711528.](#)

Initial Design/Decision Parameters



Design Capacity	Design Working Pressure	Ram Diameter	Ram Cylinder Outside Diameter	Hydraulic Lift
Decision	Decision	Design	Design	Decision
Screw Extension	Total Maximum Lift	Base Size (L*W)	Base Thickness	Operating Effort
Decision	Decision	Design/Decision	Decision/Design	Decision

Design Model (Formulae)

Physical Property	Mathematical Formula
Pressure (p)	$p = \frac{F}{a_1} = \frac{W}{a_2}$ (Pascal's Law)
Incompressibility	$a_1 x_1 = a_2 x_2$
Mechanical Advantage	$MA = \frac{W}{F} = \frac{a_2}{a_1}$
Cylinder Thickness (t) (thick cylinder)	$t = r \left[\sqrt{\left\{ \frac{\sigma_t + p}{\sigma_t - p} \right\}} - 1 \right]$ (From Lamé's Equation)
Thin cylinder (σ_θ and σ_l)	$\sigma_\theta = \frac{pd_i}{2t}$ and $\sigma_l = \frac{pd_i}{4t}$
Section Modulus (Z)	$Z = \frac{M_b}{\sigma_b} = \frac{\pi d^3}{32}$ (circular section)
Velocity ratio (VR)	$VR = \frac{\text{lift of load}}{\text{displacement of effort}}$
Breakeven quantity (BEQ)	$BEQ = \frac{FC}{SP - V_u}$

First Design Model (Assumptions)

1. Rated Capacity: $W_{rated} = 3 \text{ ton}$
2. Fluid pressure: $p = 25 \text{ MPa}$
3. Material: Mild Steel for all components (First Design)
4. Factor of safety: 5
5. Permissible Tensile Stresses: $\sigma_t = 120 \text{ MPa}$
6. Permissible shear stresses: $\tau = 45 \text{ MPa}$
7. Permissible compressive stresses: $\sigma_c = 20 \text{ MPa}$
8. Human Effort on the lever: $e = 20 \text{ kgf}$
9. Hydraulic Lift: $H_1 = 115 \text{ mm}$
10. Screw Extension: $S = 60 \text{ mm}$
11. Base Thickness: $t_b = 5 \text{ mm}$

First Design Model (Design)

Ram Diameter:

$$A_r = \frac{W}{\sigma_c} = \frac{30000}{20} \text{ mm}^2 = 1500 \text{ mm}^2 \Rightarrow d_r = 43.7 \text{ mm}$$

choose: $d_r = 45 \text{ mm}$ implies working pressure of 18.9 MPa.

Ram cylinder wall thickness:

$$t \geq \frac{pd_r}{2\sigma_t} = \frac{25 \cdot 45}{2 \cdot 120} \text{ mm} = 4.7 \text{ mm}$$

choose $t = 6 \text{ mm}$ implies outer diameter = 57 mm

Base shear check:

$$A_{sh} = \pi d_r t_b = \pi \cdot 45 \cdot 5 \text{ mm}^2 \approx 706 \text{ mm}^2$$
$$\Rightarrow \tau = \frac{W}{A_{sh}} = \frac{30000}{706} \text{ MPa} = 42.5 \text{ MPa} < 45 \text{ MPa}$$

Oil circulation volume:

$$V_0 = A_r \cdot H_1 = 1590 \cdot 115 \text{ mm}^3 \approx 183 \text{ cc}$$

Oil reservoir volume:

$$V_{res} = 1.33 \cdot V_0 = 183 \cdot 1.33 \approx 244 \text{ cc} \Rightarrow \text{a suitable choice } d_{res} = 70 \text{ mm and } h_{res} = 80 \text{ mm}$$

First Design Model (Design)

Plunger and plunger-cylinder and lever design:

Plunger diameter = 8mm (design decision)

Cylinder thickness (from the thick cylinder equation) $\approx 1 \text{ mm}$

Choice: 5mm (standard)

$$A_p = \pi * \frac{d_p^2}{4} = \pi * \frac{8^2}{4} \text{ mm}^2 \approx 50.3 \text{ mm}^2$$

Load acting on the plunger: $W_p = pA_p = 25 * 50.3 \approx 1258 \text{ N} \approx 126 \text{ kgf}$

For the human effort of 20 kgf: required VR: $VR = \frac{126-20}{20} = 5.3$ (lever rule)

Choose load arm $l_l = 5 \text{ cm}$, implies effort arm $l_e = 50 * 5.3 \text{ mm} = 265 \text{ mm}$, implies length of the lever = 270 mm.

Maximum bending moment = $265 * 200 \text{ Nmm} = 53000 \text{ Nmm}$

$$\Rightarrow Z = \frac{M_b}{\sigma_b} = \frac{53000}{120} \approx 442 \text{ mm}^3 = \frac{\pi d^3}{32} \Rightarrow d = \sqrt[3]{\left\{32 * \frac{442}{\pi}\right\}} \text{ mm} \approx 16.5 \text{ mm}$$

Choose $d = 18 \text{ mm}$

First Design Model (Design)



Checking base for compressive stress:

*Base plate dimensions = $(94 * 50 * 5) \text{ mm}^3$*

$$\sigma_c = \frac{30000}{94*50} = 6.38 \frac{N}{\text{mm}^2} \ll 20 \frac{N}{\text{mm}^2}$$

Comparison of Designs

Parameter	Rated Capacity	Design Working Pressure	Ram Diameter (d _r)	Ram-Cylinder Outer Diameter (D _c)	Hydraulic Lift (H ₁)
Present 3-ton design (paper method)	3 t	25 MPa (assumed)	45 mm	57 mm	115 mm
Taparia HBJ3 (datasheet)	3 t	≈49 MPa (back-calculated from 28 mm ram)	≈28 mm (D1)	Around 40 mm	≈115 mm

Parameter	Screw Extension (S)	Total Maximum Lift	Base Size (L × W)	Base Thickness	Estimated Operating Effort
Present 3-ton design (paper method)	60 mm	≈175 mm	94 × 55 mm	5 mm	≈ 20 kg on 270 mm lever
Taparia HBJ3 (datasheet)	≈60 mm	≈175 mm	≈94 × 55 mm	5 mm	40 kgf and 20 kgf (assuming full load 49 Mpa) on 15 cm and 30 cm levers, respectively

Calculations:



TOP PLATE

RAM ASSEMBLY

Max. mass to be lifted = 3 ton = 3000 kg

Assume, $g = 9.81 \frac{m}{s^2}$

F (Max weight to be lifted) = $3000kgf = 29.43 kN$

d (ram assembly) = 28.82 mm

$$A_1(\text{Area of ram}) = \pi * \frac{d^2}{4} = 652.35 \text{ mm}^2$$

$$\begin{aligned} \text{Pressure applied to the fluid} &= 29.43 * \frac{10^3}{652.35} \frac{N}{\text{mm}^2} \\ &= 45.1138 \frac{N}{\text{mm}^2} \end{aligned}$$

From Pascal's law same pressure is applied at the bottom of the pump plunger base.

So, pressure acts = 45.1138 MPa

$$\begin{aligned}d'(\text{base dia. of plunge pump}) &= 10\text{mm} \\A'(\text{base area}) &= \pi * \frac{d'^2}{4} \\&= 78.5 \text{ mm}^2\end{aligned}$$

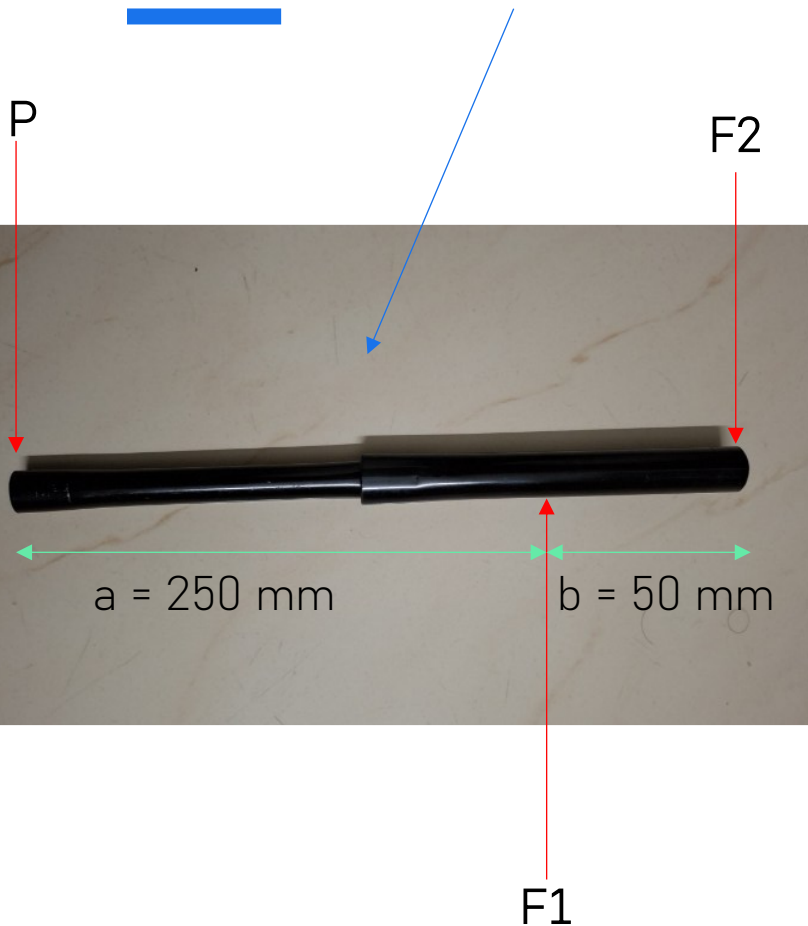
*since, pressure is same,
force applied on the plunger pump, F*

$$\begin{aligned}F &= 45.1138 * 78.5 \text{ N} \\&= 3541.43 \text{ N}\end{aligned}$$



PLUNGER

Handle Assembly



P = applied force to lift the load

$F1, F2$ = force during contact between handle assembly and plunger pump

Using moment balance equation:

*About point P , $a * F1 = (a + b) * F2$*

$$\Rightarrow 250 * F1 = 300 * F2$$

$$\Rightarrow \frac{F2}{F1} = \frac{5}{6}$$

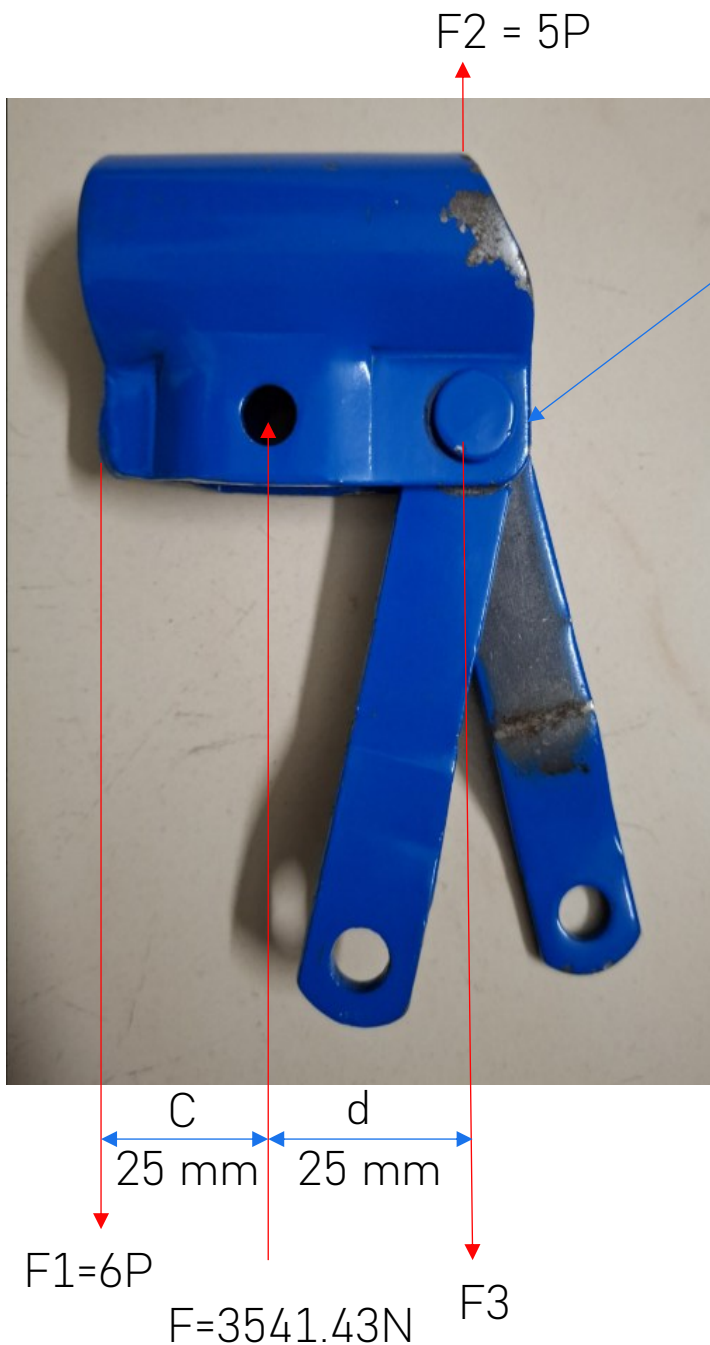
Force balance:

$$P + F2 = F1$$

$$\Rightarrow \frac{P}{F1} = \frac{1}{6}$$

$$\text{or, } F1 = 6P$$

$$\text{Also, } F2 = 5P$$



Handle Socket Assembly

Using moment balance equation:

$$\text{About point } F1, C * F = (C + d) * F3$$

$$F3 - 5P = 1770.71 \text{ N}$$

Force balance:

$$F1 + F3 = F2 + F$$

$$F3 + P = 3541.43 \text{ N}$$

$$\text{Hence, } P = 295.12 \text{ N} = 30.08 \text{ Kgf}$$

$$F1 = 180.48 \text{ kgf}$$

$$F2 = 150.4 \text{ kgf}$$

$$F3 = 330.91 \text{ kgf}$$

SPRING IN SAFETY VALVE



Material- Chrome Silicon

Type-A401

E = 29.5 Mpsi

G=11.2 Mpsi

Squared and ground

Total coils, $N_t = 7$

Active coils, $N = 5$

Using Castigliano's theorem:

$$U = \frac{T^2 L}{2GJ} + \frac{F^2 L}{2AG}$$

$$k = \frac{d^4 G}{8D^3 N} = \frac{209.043 \text{ N}}{\text{mm}} (\text{very high stiffness})$$

Features of this spring:

1. maintains uniform pressure
2. reduced *buckling*
- 3 provides stability

Base of hydraulic jack

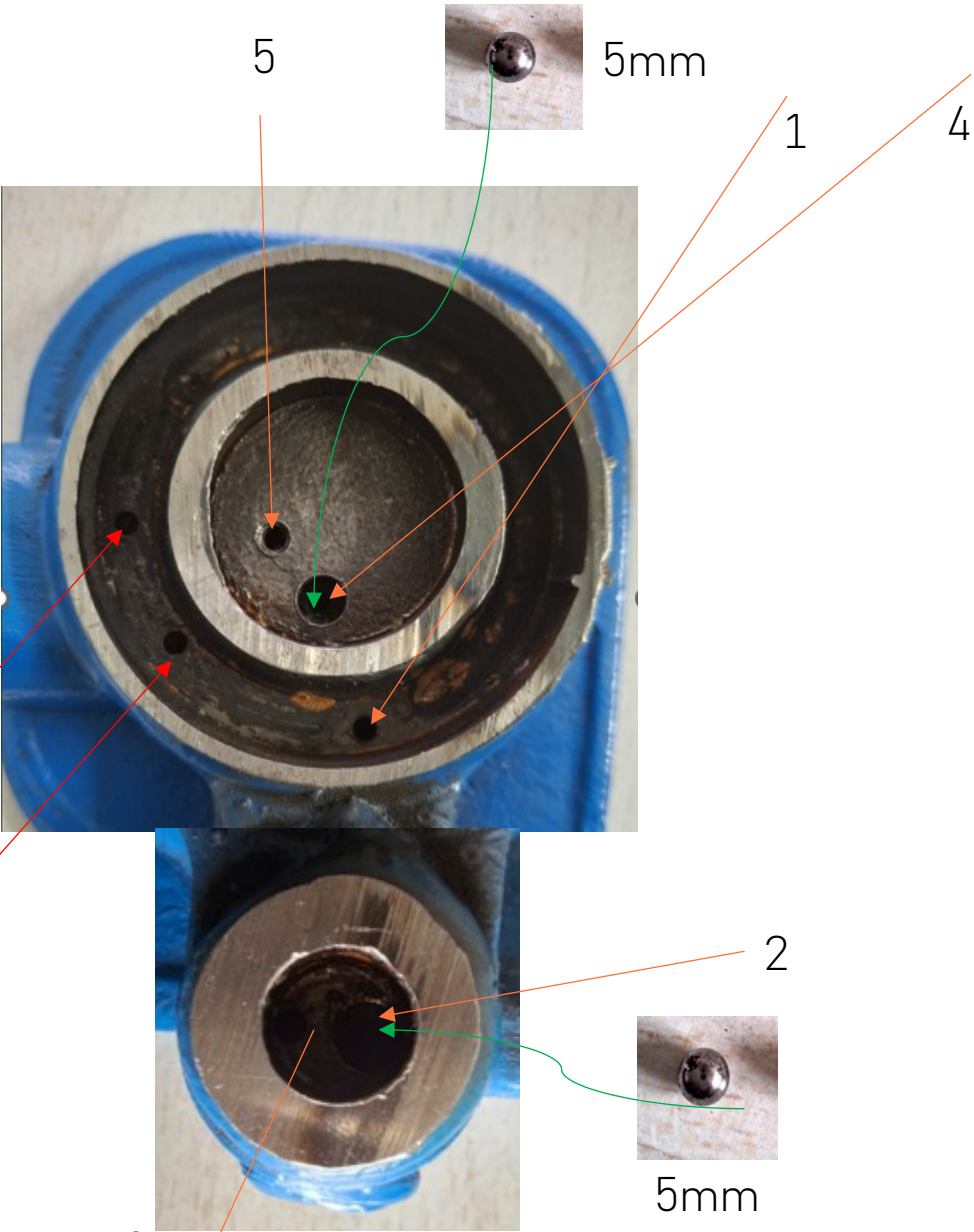
Plunger moves up
=>oil reservoir to
plunger pump (1-2)

plunger moves down
=>oil from plunger
pump to ram area
lifts the load. (3-4)

Release valve

Safety valve

3
small opening



Release valve



Safety valve



Justifications for the Jack Material

Component	Specified Material	Key Property (Strength/Resistance)	Operational Justification ("So What?")
Cylinder Sleeve	Steel (H8 Ra 0.4)	Internal smoothness; High tensile strength	Mandated to prevent fluid bypass and ensure seal longevity under 500 bar working pressure.
Ram and Plunger	Chromium-plated C45 Steel	High tensile strength; Wear resistance	C45 upgrade ensures the safety factor for 3-tonne loads; plating prevents rod scoring.
Ram (Piston part)	Steel (CD-DK Series)	High compressive strength	Required for 3-tonne 500 bar applications to prevent deformation.
Seals	Polyurethane / NBR	Temp Resilience (-10°C to +80°C)	Ensures leak-proof operation in freezing roadside conditions and compatibility with mineral oil.

Justifications Continued

Component	Specified Material	Key Property (Strength/Resistance)	Operational Justification ("So What?")
Bushings	Sintered Bronze	Self-lubricating properties	Reduces mechanical friction to prevent rod damage and subsequent catastrophic bypass leakage.
Fasteners	Quenched Steel (Galvanized)	High yield strength; Rust resistance	Maintains structural assembly integrity and prevents fastener failure from oxidative stress.
Handle/Arm	Mild Steel	250 MPa Yield Strength	Provides the necessary structural margin for load transmission at the lowest weight-to-cost ratio.
Base Plate	Ductile Iron Based	Impact and shock resistance	Prevents catastrophic brittle failure during eccentric loading common in roadside lifting.

A Few Material Improvements

COMPONENT	INFERIOR MATERIAL/CONSTRUCTION	SUPERIOR MATERIAL/CONSTRUCTION	RATIONALE FOR SUPERIORITY
Base Plate	Thin, stamped steel or low-grade cast iron	Forged steel or thick, high-grade ductile iron	Forging aligns grain structure for maximum strength; prevents bending and cracking under load.
Main Ram	Untreated, low-carbon steel	Heat-treated, high-carbon steel with hard chrome plating	Provides high compressive strength to prevent bending and a smooth, corrosion-resistant surface for seals.
Housing	Brittle cast iron	Welded steel construction or high-grade ductile iron	Welded steel offers strength and consistency; ductile iron can deform before fracturing, offering a safety margin.
Welds	Uneven, porous, with spatter	Uniform, continuous, and clean (often robotic)	A high-quality weld is as strong as the parent metal, eliminating weak points and ensuring pressure containment.

Oil Selection

•**Viscosity:** The oil must be thin enough to flow easily through the small valves and passages in the system, but thick enough to provide a good seal and lubricate the moving parts. Viscosity is also affected by temperature. A good quality jack oil will have a stable viscosity across a wide range of operating temperatures, which is vital for users working in the extreme heat of the Middle East or the cold of a Russian winter. Using the wrong fluid, such as motor oil or brake fluid, can damage seals and cause the jack to operate poorly or fail.

•**Anti-Foaming and Anti-Wear Additives:** Quality hydraulic oils contain additives that prevent the oil from foaming as it is rapidly moved through the system (air bubbles in the fluid make the jack feel "spongy" and less effective) and additives that reduce wear on the finely machined metal components.

Oil Selection



Common types of Hydraulic Oils are:

ISO VG 32 AW (Group II) AW implies anti-wear.

ISO VG 46 AW (Group I)

Plain Mineral Oil

Turbine Oil (R&O)

The most common of these for good hydraulic jacks is ISO VG 32 AW.

Reasons:

- Viscosity is typically in the range **15–68 cSt at 40°C**
 - ISO VG 32 (~32 cSt) sits right in the middle → ideal balance between
 - If viscosity is too high → hard to pump
 - If too low → leakage + poor efficiency
- Anti-wear protection (critical for jacks)
 - High pressure
 - Boundary/mixed lubrication conditions.
 - That's why **AW (anti-wear) oils** are used instead of plain mineral oil.

Oil Selection



Common types of Hydraulic Oils are:

ISO VG 32 AW (Group II) AW implies anti-wear.

ISO VG 46 AW (Group I)

Plain Mineral Oil

Turbine Oil (R&O)

The most common of these for good hydraulic jacks is ISO VG 32 AW.

Reasons:

- Oxidation stability (long life)
 - Plain mineral oil: ~300 hours
 - AW hydraulic oil (Group I): ~2000 hours
 - AW hydraulic oil (Group II): ~5000 hours

Hence, ISO VG 32 anti-wear hydraulic oil (AW-32/HM-32) is the best choice.

However, ISO VG 46 (for warmer climates or heavy-duty use).

Oil Selection: Few Equations

For simplified analysis, a very crude model is adopted.

1. Volumetric Efficiency η_v

Leakage decreases as viscosity increases:

$$\eta_v = 1 - \frac{Q_{leak}}{Q_{ideal}}$$

Leakage flow:

$$Q_{leak} \propto \frac{1}{Z}$$

So:

$$\eta_v \uparrow \text{ with } Z$$

A practical model:

$$\eta_v = 1 - e^{-k_1 Z}$$

Oil Selection: Few Equations

2. Mechanical Efficiency η_m

Viscous drag increases with viscosity:

$$T_{loss} \propto Z$$

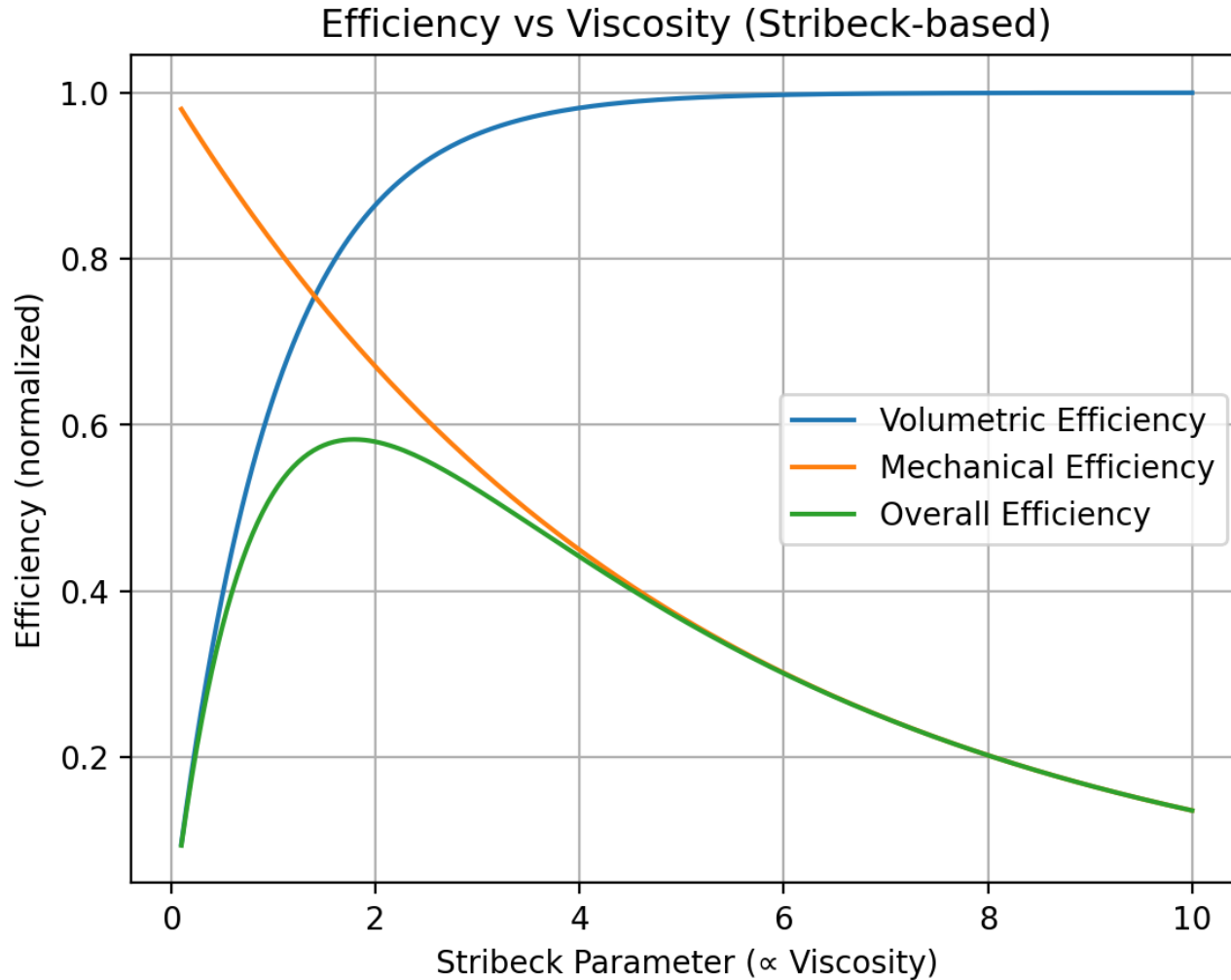
$$\eta_m = \frac{T_{useful}}{T_{input}} \approx e^{-k_2 Z}$$

3. Overall Efficiency η_o

$$\eta_o = \eta_v \times \eta_m$$

$$\eta_o = (1 - e^{-k_1 Z}) \cdot e^{-k_2 Z}$$

Oil Selection: Stribeck Curve



Oil Selection: A comparison

Factor	ISO VG 32 AW (Group II)	ISO VG 46 AW (Group I)	Plain Mineral Oil	Turbine Oil (R&O)
Viscosity @ 40°C (cSt)	~32 (ideal mid-range)	~46 (higher → harder pumping)	Variable	~32–46
Cold start performance	Good (flows easily)	Moderate (stiffer)	Poor	Good
Pump effort (manual jack)	Low → smooth operation	Higher → more force needed	Inconsistent	Moderate
Volumetric efficiency	High (good sealing, low leakage)	Slightly higher sealing but flow losses	Poor	Moderate
Mechanical efficiency	High (balanced viscosity)	Lower (viscous drag)	Poor	Moderate
Lubrication regime suitability	Excellent (hydrodynamic + boundary)	Good	Poor	Weak boundary protection, Good in HD
Antiwear protection	Strong (additives present)	Strong	None	Weak
Oxidation life (ASTM D943)	~5000 h	~2000 h	~300 h	>10000 h
Thermal stability	High	Moderate	Low	Very high
Sludge/varnish formation	Low	Moderate	High	Low (but poor wear control)
Seal compatibility	Good	Good	Variable	Good
Additive performance	Balanced (AW + antioxidants)	Moderate	None	Oxidation-focused only
Cost vs performance	Optimal	Moderate	Cheap but unsuitable	Expensive overkill
Suitability for bottle jack	Best	Acceptable	Not suitable	Not suitable

Leakage Resistance: A Solution

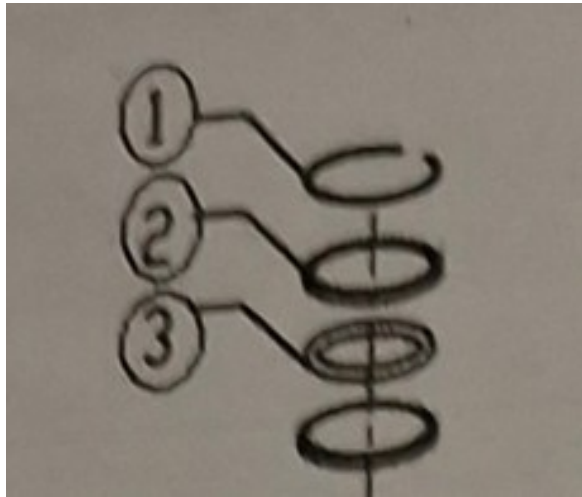
Part No	Description	Qty
1	Round wire snap ring	1
2	Ring (Rectangular Section)	2
3	O-Ring	1
4	Ram assembly	1
5	Bowl washer	1
6	O-Ring	1
7	Round wire snap ring	1
8	Base welding parts	1
9	Oil plug	1
10	Steel ball 6 mm	1
11	Release valve assembly	1
12	Plastic plug	1
13	Safety valve	1
14	Spring	1
15	Spring holder	1
16	Steel ball 3 mm	1
17	Steel ball 5 mm	2
18	Bowl washer	1
19	Pump plunger assembly	1
20	Reusable cotter pin 01.2 x 18 mm	2
21	Pin	2
22	Handle socket assembly	1
23	handle assembly	1
24	Seal kit	1

Good, but O-rings are not very helpful in reducing leakage.

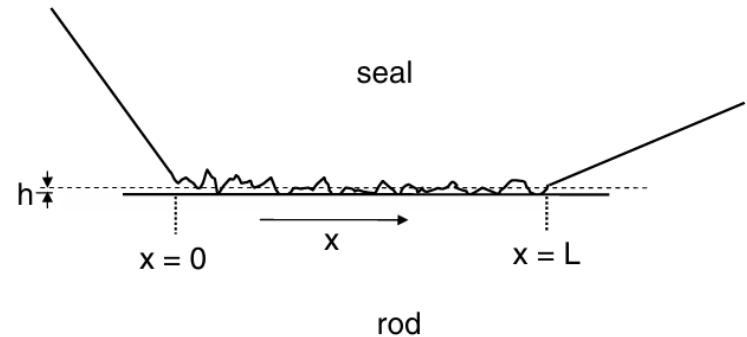
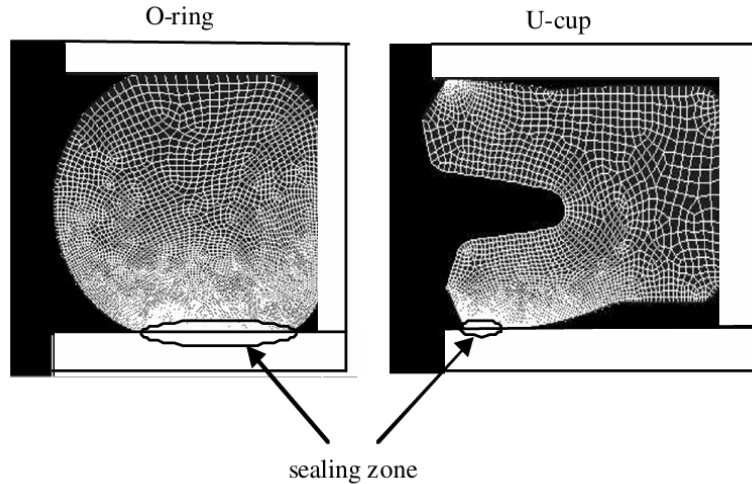
Leakage Resistance: A Solution

External Leaks: Often caused by worn "wiper" seals at the top of the cylinder. You will see fluid weeping down the side of the ram.

Internal Leaks (Bypassing): If the high-pressure seal on the piston fails, fluid leaks back into the reservoir during the power stroke. The jack might lift a load initially but will slowly "drift" or settle downward even when the release valve is closed.

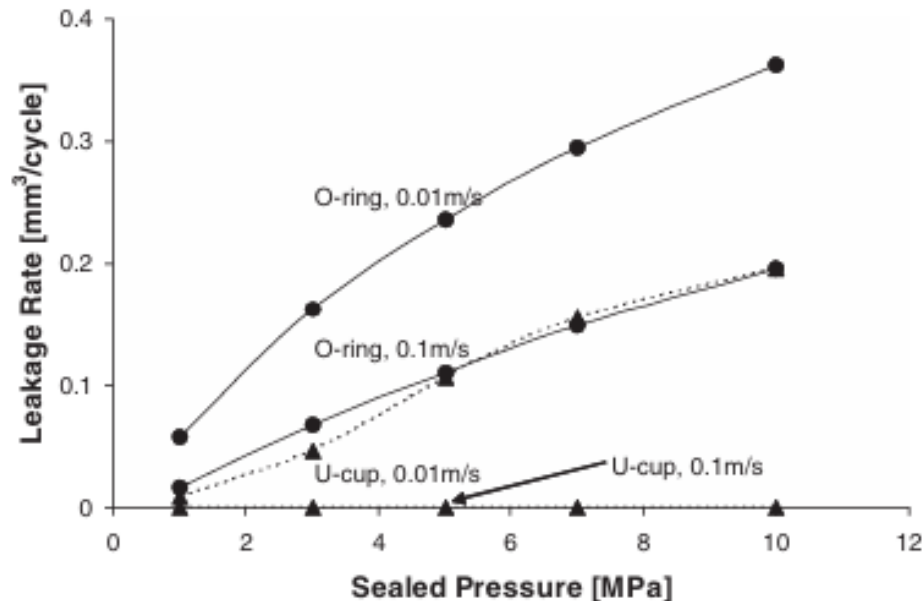


Leakage Resistance: A Solution



U-cup seals are pressure-energised, i.e., the higher the pressure, the less the leakage. When the high-pressure fluid enters the gap between the lips of the seal, it extends the lips radially, i.e., exerts more pressure on the wall, i.e., improves contact, i.e., reduces leakage.

Leakage Resistance: A Solution



These curves can be obtained using Reynold's model of flow past the seals.

Elastohydrodynamic lubrication theory also indicates that the resistance offered by U-cup seals is lower than that of O-ring seals.

Leakage Resistance: Challenges



The challenge remains in the manufacturing of U-cup seals:

These seals must be properly designed; otherwise, manufacturing or design mistakes may lead to premature failure.

For example, a smaller lip size may allow leakage, whereas a larger lip size may pull the seal out, leading to failure.

Also, if lip sizes are kept unequal, it is essential to analyse how unequal they should be; otherwise, this may lead to a single lip behaviour.

Due to these challenges, O-ring seals remain popular. They will not be pulled out of the groove due to the fluid pressure, at least.

Operational Troubleshooting

Sr. No.	Jack will not Lift Load	Jack will not Hold Load	Jack will not Lower	Poor jack Lifting	Will not Lift to Full Extension	Causes and Solutions
1			0		0	Release valve is not completely closed. (Turn handle clockwise)
2		0	0	0	0	Weight Capacity Exceeded.
3		0	0		0	Air is in the hydraulics. Purge air from system.
4			0			Low oil level. Add oil as required.
5	0	0		0	0	Oil reservoir is overfilled. Drain excessive oil. Lubricate moving parts.
6	0	0		0	0	Jack is binding or there is foreign particles obstruction.
7			0			Power unit malfunctioning. Replace the power unit.

References

1. Paul W. Michael¹ and Hongmei Zhao (2019) "Fuels and Lubricants Handbook: Technology, Properties, Performance, and Testing", chapter 17
2. K.Sainath, Mohd. Salahuddin, MdAzam Ali Farooky, Mohammed Siddique Ahmed, Mohd. Riyaz Uddin, Faraz Ur Rehman Azhar, Md Shaffi (2014) "Design of Mechanical Hydraulic Jack".
3. Muhammad Harith Roszaidi¹, Wan Mohd Wardi (2023) "Simulation Analysis for Big Piston of Hydraulic Bottle Jack"
4. B Yang and R Fsalant (2010) "Elastohydrodynamic lubrication simulation of O-ring and U-cup hydraulic seals"
5. Richard G. Budynas, J. Keith Nisbett "Shigley's Mechanical Engineering Design", tenth edition
6. Website: [An Expert's Guide: 5 Key Checks Before Buying a 20 Ton Bottle Jack in 2025 - TOYO Heavy Industries](https://www.toyo-industry.com/) (<https://www.toyo-industry.com/>)
7. YouTube: <https://youtu.be/TMhrBeGkYow?si=PBsEpNh6ggFQ6B7G>